

Unsupervised Learning Assignment

General Guidelines for Assignment for Unsupervised Learning Component

You must submit your solutions as a Jupyter Notebook for this assignment in both .html and .ipynb formats, as well as a .csv file that will be called `electrophysiology_data_kmeans.csv`. The HTML file will be reviewed for readability, the ipynb file will be used to verify solutions, and the CSV file will be checked for format and consistency with your notebook (cluster numbers are arbitrary, so there is no single correct file to compare with).

The following instructions outline the key components and guidelines for preparing the notebook:

Notebook Structure and Content

1. Headers and Subheaders:

- Organize the notebook into clearly defined sections using headers and subheaders.

2. Graphs and Visualizations:

- Include well-labeled graphs with appropriate titles, x and y-axis labels, and scales.
- Specify the colors used in each graph to ensure clarity and consistency.
- Each graph also includes a subsequent markdown cell to explain the graph's purpose and relevance to the analysis.

3. Work individually or in pairs:

We will contact each student to explain the work done individually or in pairs with questions to verify that the student understands the solution submitted in depth. We will only accept assignments submitted jointly by up to two students or submitted individually. The meeting will take place online (Google Meet, about 15 minutes per student). You may use AI assistants (for

example ChatGPT, Claude or Copilot) to help you write code, but you are responsible for understanding every step: in the meeting we will ask how you solved the task, why you chose your methods, what your results mean and what their limitations are. We grade this understanding, not the code.

4. Grading:

Most of the grading in this section will be on evidence of applied learning in the course. Explain the reasoning behind the methods you selected. Provide visual evidence to confirm the explanations. Achieving a high level of success on performance measures may yield additional evidence of correct thought processes; however, success in performance measures is less significant in grading. Using tools you know or found online that are beyond what was taught in this course is acceptable; however, if you use such tools, please explain them at a level that a freshman majoring in engineering or hard sciences could follow, as we want to follow what you are doing as well. Also, we will ask you verbally about the components of your solutions, so be prepared to demonstrate command of the tools you used. Knowing what your analysis shows, what it does not show and how sure you can be is as important as running the methods: a modest result that you understand and report honestly is worth more than an impressive-looking picture that you cannot explain.

5. Submission (what to upload):

- Three files, uploaded together: (1) your notebook, UnsupervisedLearning_<your name(s)>.ipynb, saved WITH its outputs; (2) the same notebook exported to HTML; (3) electrophysiology_data_kmeans.csv (62 rows: the 8 feature columns plus the column kmeans_labels, without the unnamed index column).

- Before saving and exporting, run Kernel > Restart & Run All once, so that the notebook runs from the first cell to the last and the saved outputs match the code.
- Do not upload the data file. If you worked in pairs, put both names in the first cell and submit once.
- Upload to: [the submission form](#). You need to be signed in to a Google account to upload files. Deadline: November 17, 2026. Then book your online meeting (Google Meet, November 25–26, 2026) by November 20 on [the booking page](#). If you worked in a pair, each partner books their own meeting.

The Data

1. The unsupervised learning component of the homework assignment involves analyzing a dataset structured for behavioral analysis of subject rats in dyadic social experiments. The dataset consists of 8 columns representing some of the features extracted from electrophysiological data refined for this assignment by performing data imputation and cleaning. The features are extracted from electrophysiological measurements collected from brain areas important in mediating social behavior. The data set has 62 samples (rows). The features are anonymous (feature_1 to feature_8) and there are no labels. The csv file also contains an unnamed first column with the row number; it is not a feature, so remove it before your analysis.
2. Link to data: <https://github.com/deutschlab/computational-ethology-final-project/tree/main/unsupervised/data>. Course repository with a starter notebook: <https://github.com/deutschlab/computational-ethology-final-project>

Assignment Tasks

Preprocessing (all tasks): remove the unnamed index column and work with the standardized features (mean 0, standard deviation 1; see item 3), unless an item asks you to compare with the unscaled data.

1. **Dimensionality reduction (50 points):**
 - Reduce the dimensions of the data to 2 dimensions using t-SNE, UMAP, and PCA
 - Perform the reduction using the following parameters and visualize each reduction:
 1. t_SNE: perplexity level: [5, 50] (2 plots) - perplexity must be smaller than the number of samples (62)
 2. UMAP: n_neighbors: [2, 10], min_dist: [0.1, 1.0] (4 plots; min_dist cannot be larger than the UMAP parameter spread, whose default value is 1)
 3. PCA (default) (1 plot)

- Visualize the data for each reduction using a scatter plot
- Explain how the different parameters affect the reduction (compare your plots: what stays similar across the settings and what changes?)
- Also try t-SNE with perplexity 100 and UMAP with min_dist 10. Read the error messages and explain, in your own words, why these settings do not work for this data set.

2. Clustering (50 points):

- Cluster the data using KMeans, AgglomerativeClustering, and one more method of your choosing (for example Gaussian Mixture or DBSCAN); explain why you chose it and how you set its parameters
- For Kmeans clustering, evaluate the optimal number of clusters using the Elbow method
 1. Fit 10 Kmean models with the number of clusters ranging from 1-10
 2. Extract the inertia of each fitted model
 3. Plot the Elbow plot
- Add a column to the data frame named "kmeans_labels" and assign the labels extracted using the K-means method. Use the number of clusters that you chose (state it and explain your choice). Save the data frame, without the unnamed index column and with the new column, as electrophysiology_data_kmeans.csv.

3. Feature Scaling ([TBD] points):

- Standardize the 8 features (mean 0, standard deviation 1) before running PCA, t-SNE, UMAP and the clustering.
- Compare the PCA plot with and without scaling. Does the picture change?
- Also compare the elbow plot with and without scaling. Which one looks clearer, and is the clearer one also the more trustworthy? Explain.

Why we ask: *without scaling, the feature with the largest numbers dominates every distance and every principal component.*

4. Silhouette Score ([TBD] points):

- For the KMeans models with 2 to 10 clusters, compute the silhouette score and plot it against the number of clusters, next to the elbow plot.

- Do the two methods agree on the number of clusters?
- Repeat the KMeans fits with a different random_state. Does the best number of clusters change? What does that tell you about how much you can trust the choice of k? (It is a valid result to conclude that the data do not show one clear number of clusters.)

Why we ask: *the silhouette score is a number describing how well each point fits its own cluster compared with the nearest other cluster (higher is better). It is useful because the bend of the elbow plot is not always obvious.*

5. Cluster Profiles ([TBD] points):

- After choosing the number of clusters, show the mean of each original feature per cluster (a table or a bar plot).
- Because the features are on very different scales, compare standardized means (for example in a heatmap), and report how many samples each cluster contains.
- Describe each cluster in one sentence using the original features.
- Are there very small clusters (2-3 samples)? What could they be?

Why we ask: *looking at the original features per cluster tells you what the clusters are beyond their color.*

6. Are the Clusters Real? ([TBD] points):

- Create a control data set by shuffling every column of your standardized data independently (this keeps each feature's values but destroys the relationships between features). Run KMeans on it with the same number of clusters, compute the silhouette score, and repeat a few times.
- Compare these scores with the score of the real data. What do you conclude about the clusters in the real data? There is no single right answer: explain what this comparison does and does not show.

Why we ask: *a clustering algorithm always returns clusters, even for data without structure. A control shows what a "no structure" result looks like.*

7. Summary: Strengths and Limitations ([TBD] points):

- Write a short summary (about 6-10 sentences, in a markdown cell) that answers: (a) what structure, if any, did you find in the data, and how confident are you; (b) which of your results depend on your choices (scaling, method, number of clusters, random seed) and which are stable; (c) what would you need (more samples, labels, knowing what the features are) to be sure?

Why we ask: *this is the part that shows that you understand your work and not only that you ran it.*